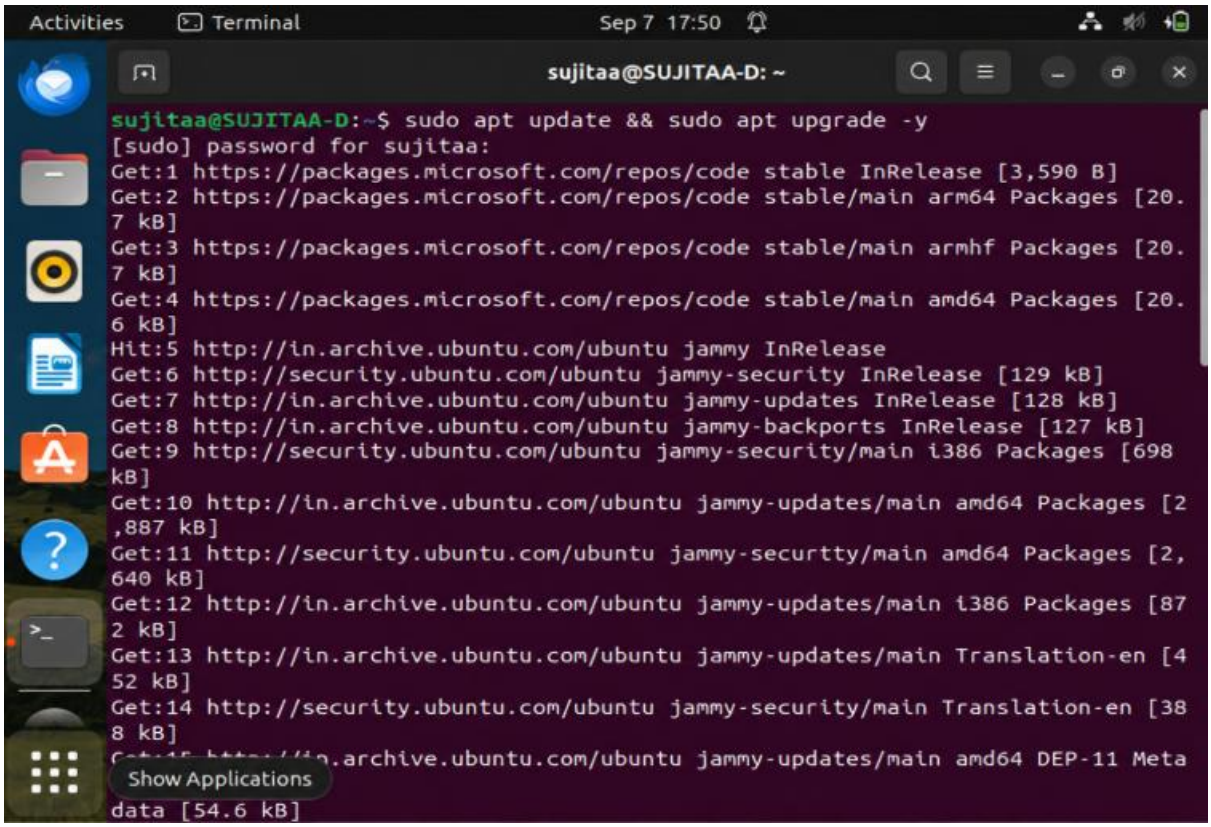
9) Installation of OpenStack and creation of Virtual machines

Aim:

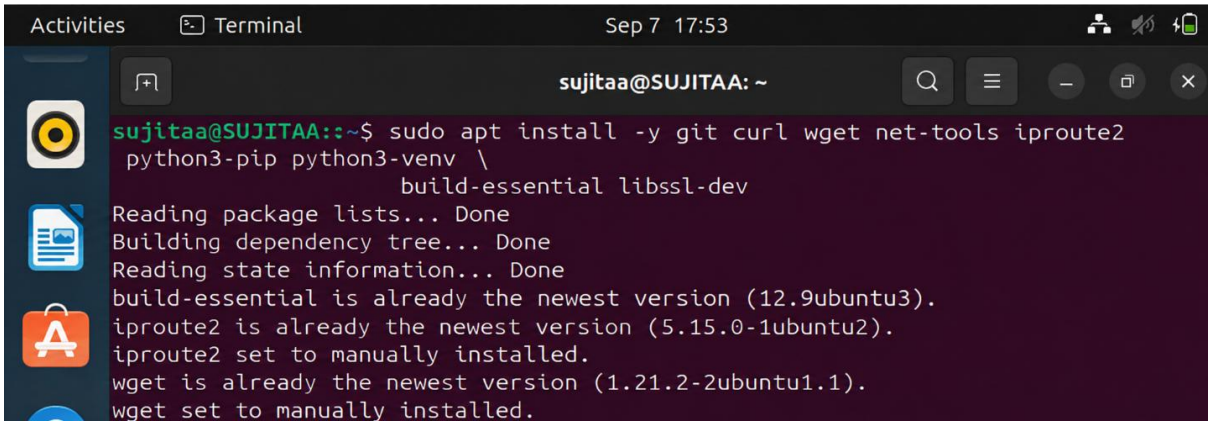
To install of openstack and creation of virtual machine in linux system.

Steps:

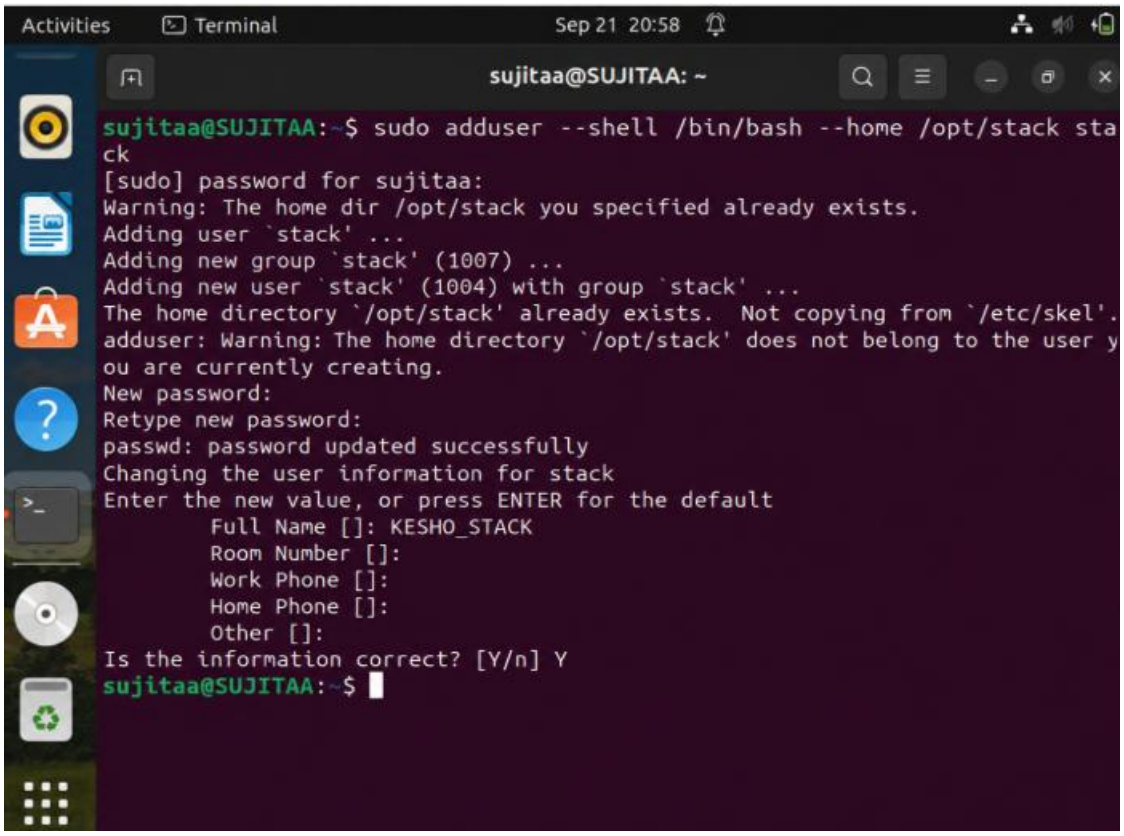
1)Update all the file in the ubunttu by sudo apt update



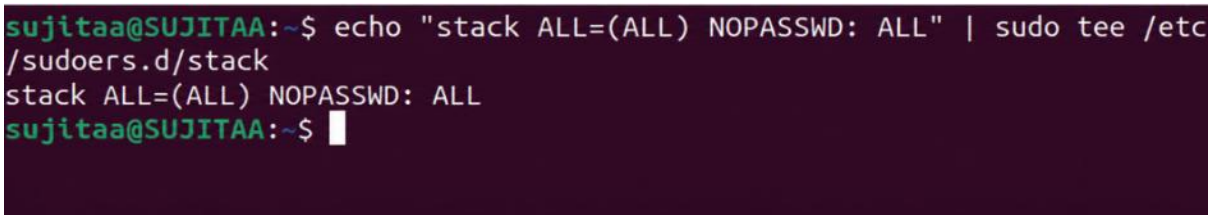
2)Install the packages to build the openstack



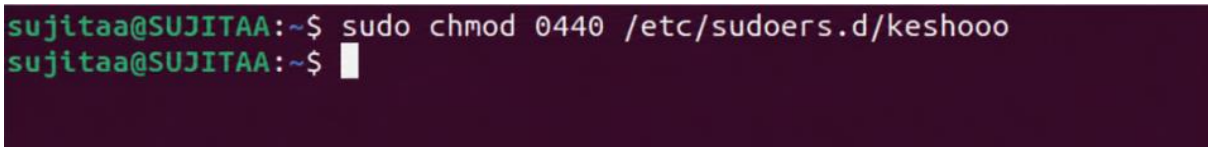
3) Add the user stack to the ubunttu especially for the openstack and give the sudo privilege for the user stack and add to the group



4) Set no password for all



5)Give the privileges for the user stack



6) open the keshoo as sudo user by the command sudo su - stack

```
stack@sujitaa:~$ git clone https://git.openstack.org/openstack/devstack
Cloning into 'devstack'...
warning: redirecting to https://opendev.org/openstack/devstack/
remote: Enumerating objects: 52016, done.
remote: Counting objects: 100% (31604/31604), done.
remote: Compressing objects: 100% (10731/10731), done.
Receiving objects:   0% (292/52016), 108.00 KiB | 24.00 KiB/s
```

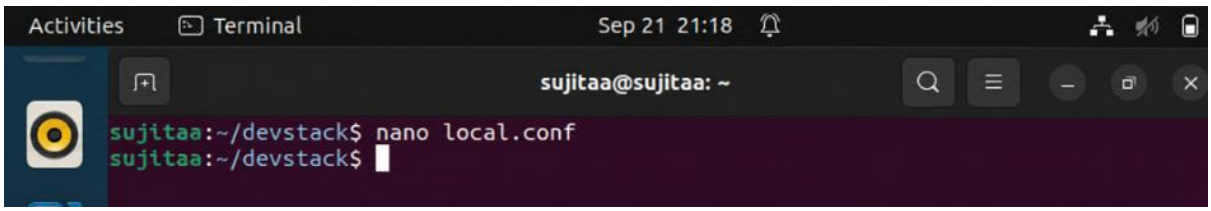
7)go to home directory and download the openstack from the git by cloning the git .


```
stack@sujitaa:~$ git clone https://git.openstack.org/openstack/devstack
Cloning into 'devstack'...
warning: redirecting to https://opendev.org/openstack/devstack/
remote: Enumerating objects: 52016, done.
remote: Counting objects: 100% (31604/31604), done.
remote: Compressing objects: 100% (10731/10731), done.
Receiving objects:   0% (292/52016), 108.00 KiB | 24.00 KiB/s
```


8) Move to the devstack directory

```
sujitaa@SUJITAA:~$ cd devstack
sujitaa@SUJITAA:~/devstack$
```

9)Change the configuration file in the local.conf



Update the below line and give password for each

```
[[local|localrc]]

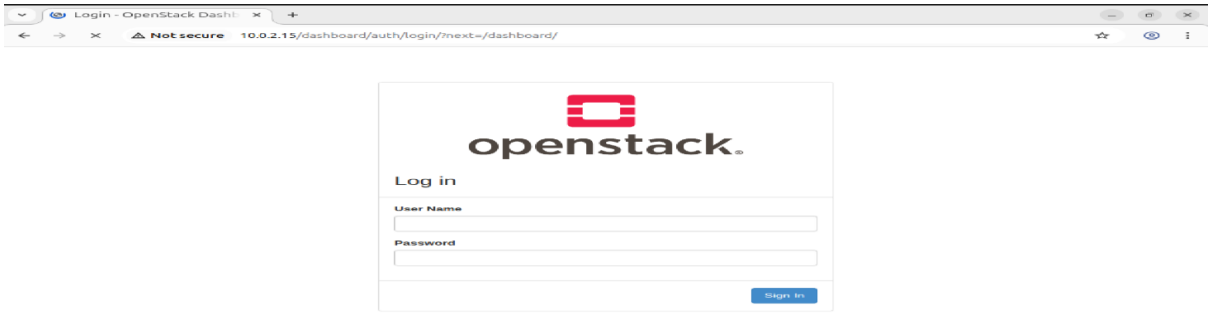
ADMIN_PASSWORD=dtuk2004
DATABASE_PASSWORD=$ADMIN_PASSWORD
RABBIT_PASSWORD=$ADMIN_PASSWORD
SERVICE_PASSWORD=$ADMIN_PASSWORD

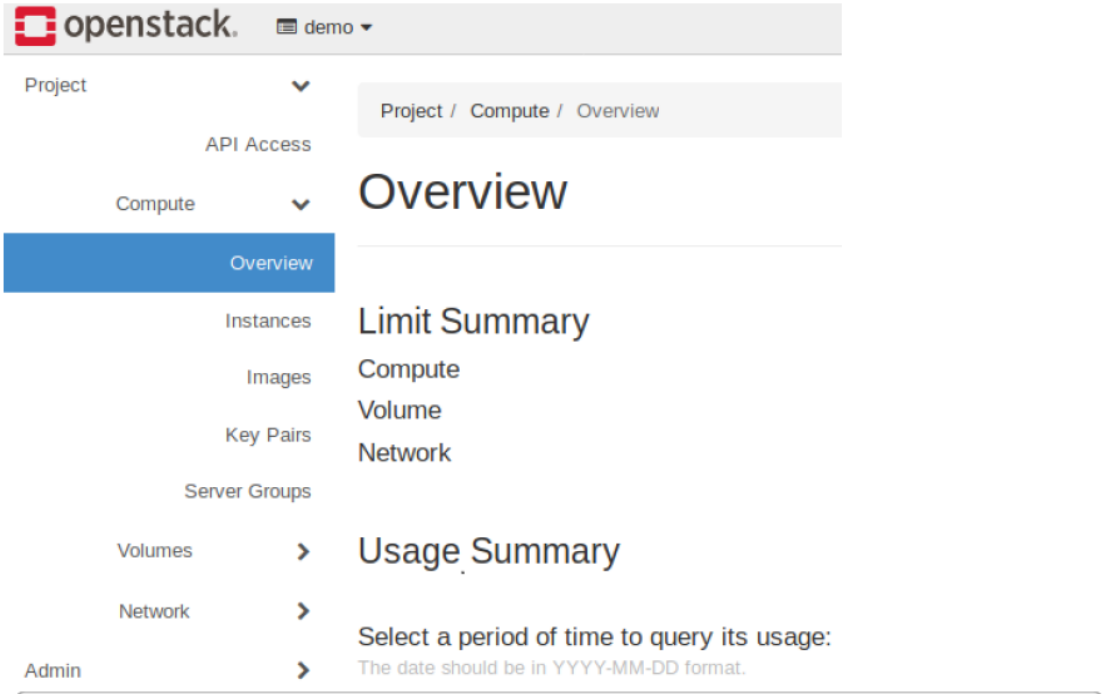
DEST=/opt/stack
HOST_IP=192.168.1.10
```

10)Run the openstack by the ./stack.sh

```
stack@KESHOKUMAR-D:~/devstack$ ./stack.sh
+ unset GREP_OPTIONS
+ unset LANG
+ unset LANGUAGE
+ LC_ALL=en_US.utf8
+ export LC_ALL
++ env
++ grep -E '^OS_'
++ cut -d = -f 1
+ unset
+ umask 022
+ PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:
/usr/local/games:/snap/bin:/usr/local/bin:/usr/local/sbin:/usr/sbin:/sbin
+++ dirname ./stack.sh
++ cd .
++ pwd
+ TOP_DIR=/opt/stack/devstack
+ NOUNSET=
+ [[ -n '' ]]
```

11) Visit the IP address that you have configured in the local.conf and login with the password configured in the local.conf





12) Create a Virtual Machine using openstack glace service using the following command

a) Source OpenStack Credentials.

source devstack/openrc admin admin

(Replace admin with your actual project and username if different). Verify Image and Flavor.

b) Checkout the image list

openstack image list

openstack flavor list

Note the name or ID of the image and flavor you intend to use.

c) Create a Key Pair (if not already done):

openstack keypair create mykey > mykey.pem

chmod 600 mykey.pem

d) Create a Network

DevStack typically creates a default network. You can list existing networks:

openstack network list. If you need to create a new one, this involves creating a network, subnet, and router, and connecting them. Launch the Virtual Machine.

openstack server create --image <image_name_or_id> --flavor <flavor_name_or_id> --nic net-id=<network_id> --key-name mykey my-first-vm

Replace <image_name_or_id>, <flavor_name_or_id>, and <network_id> with your chosen values. my-first-vm is the desired name for your VM. Verify VM Status and IP Address.

openstack server list

Look for your VM and its assigned IP address. Access the VM.

ssh -i mykey.pem cirros@<vm_public_ip>

Result:

Thus the installation of openstack and creation of virtual machine in linux system has been executed successfully and output has been verified.